



BUDDING PARENT

An adult hydra can produce young asexually by developing small buds, which grow into new individuals. This parent has grown a single bud, which is now several days old. Eventually, the young animal will break away and become independent.

Reproduction

Invertebrates have a wide range of reproductive techniques, and include animals that reproduce both asexually and sexually. Asexual reproduction usually involves just a single parent. In sexual reproduction, 2 separate individuals are normally needed, although sometimes a single animal can simultaneously take on both male and female roles.

Few invertebrates rely solely on asexual reproduction, because it can lead to problems by creating young that are genetically identical to one another and their parents (see p.30). However, for many species—particularly sap-sucking insects, such as aphids—asexual reproduction is a valuable way of boosting numbers when food is in plentiful supply. Internal parasites, such as flukes, also use asexual reproduction at certain stages of their life cycle. For them, this way of multiplying is particularly important, because their complex life histories mean that each individual offspring has a very small chance of survival. However, with almost all these animals, sexual reproduction intervenes at some point in the life cycle, creating the genetic variations that allow a species to adapt to changing conditions around them.



NEW LIFE

Released by a coral polyp, a parcel of eggs and sperm floats upward. At the sea surface, the parcel will separate, the eggs and sperm will mingle with those of other individuals, and fertilization will occur.

Although asexual reproduction is widespread among invertebrates, most species use sexual reproduction alone. With vertebrates, sexual reproduction almost always involves a male and female animal pairing up, but in the invertebrate world things are often quite different. Animals with both male and female reproductive organs—known as hermaphrodites—are common, and these may either pair up, as in the case of earthworms, or they may fertilize themselves.

Even where individual animals are either male or female, they may reproduce without ever having to meet. This apparently paradoxical situation is common among marine invertebrates, such as sea urchins, and it happens because these animals reproduce by external fertilization. They release their sex cells into the water, and the sperm and egg cells mingle, producing large numbers of fertilized eggs. For this form of reproduction to succeed, the eggs and sperm must be released with precise timing, which is often achieved by coordinating with the changing phases of the moon.

On land, most invertebrates use internal fertilization, with the male and female pairing up, and the male inserting his sperm into the female's body. This can involve complex courtship displays, and sometimes—for example, in spiders—careful behavior on the part of the male, who runs the risk of being eaten by his mate. In the insect world, some male bugs adopt a less ceremonious approach: they use “traumatic insemination,” piercing the female's abdomen with an organ that works like a dart. However, internal fertilization does not necessarily involve the direct transfer of sperm. In some scorpions, for example, the male produces a packet of sperm, known as a spermatophore, which it fixes to the ground. The male then presses the female onto the spermatophore, forcing it into her internal genital chamber, where fertilization takes place.

Parental care

Once a female's eggs have been fertilized, most invertebrates leave their young to develop on their own. As a result, their young face an uphill struggle during their first few weeks of life, and a large proportion fail to survive. These losses are offset by the large number of eggs that are produced. A female starfish, for example, can release 2.5 million eggs in 2 hours.

At the other extreme, some invertebrates are attentive parents. Female earwigs, for example, remain with their eggs, periodically cleaning

METAMORPHOSIS

Most invertebrates change shape as they grow up. This change, known as metamorphosis, allows the young and adults to live in different ways, which can improve their chances of finding food.

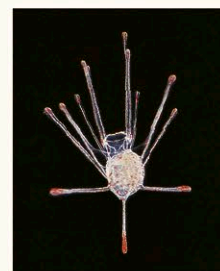
It also helps species to disperse, because one stage—either the larva (juvenile) or the adult—can have adaptations that enable it to travel. Metamorphosis may be gradual, or rapid and abrupt. In the first type, the young often resemble their parents, but in the second, they look completely different. Metamorphosis is controlled by hormones.



EARLY STAGE

DRIFTING YOUNG

Sea-urchin larvae have slender spikes that help them to drift in the plankton. As a larva matures, the spikes are slowly absorbed.



LATE STAGE



pupal case
wings showing through

COMPLETE CHANGE

During the pupa stage, a caterpillar's body is broken down and an adult butterfly is formed in its place. The change occurs inside a chrysalis, or pupal case.



SAFE PASSAGE

With her family crowded on her back, a female scorpion guards her brood. At this stage, the young have pale, soft exoskeletons—these darken and harden as they age.

them to ensure that they are not attacked by parasites or by mold. Female octopuses attach their eggs to rocky recesses on the seabed, and remain nearby, squirting jets of water over them to keep them clean and well oxygenated. Invertebrates may also take care of their young after

they have hatched. Many arachnids carry their young on their back, while some freshwater crayfish carry theirs on their legs.

Colonies

Even if they are cared for, most young invertebrates soon disperse to take up life on their own. But with colonial species, animals from the same species remain together, forming permanent groups or colonies. In a colony, the members are usually closely related to one another, and they often divide up the work needed for survival.

Colonial life is a recurring theme in invertebrate evolution, and it can take a variety of different forms. In some colonies—particularly in the sea—the members are physically joined, and may be so highly integrated that they look and behave like a single animal. This kind of organization is seen in

COLONY AFLOAT

A Portuguese man-of-war is an animal colony, formed by a collection of specialized polyps that live and work together.

